

Experiment 6

Question

Perform basic video processing operations on the captured video. Read the captured video in Python and display the video in slow motion and fast motion.

Aim

To read a video using Python and OpenCV and display it in slow motion and fast motion.

Procedure

1. Import the OpenCV (cv2) library.
2. Read the video using cv2.VideoCapture().
3. Check whether the video is opened successfully.
4. Read each frame of the video using a loop.
5. Display the video frame by frame.
6. Adjust the playback speed using cv2.waitKey().
7. Use cv2.waitKey(100) for slow motion & cv2.waitKey(10) for fast motion.
8. Press 'q' to stop video. And Save Output Video

Code (Slow Motion)

```
import cv2

video = cv2.VideoCapture("sample.mp4")

if not video.isOpened():
    print("Video not found!")
else:
    while True:
        ret, frame = video.read()

        if not ret:
            break

        # Resize frame
        frame = cv2.resize(frame, (640, 480))

        cv2.imshow("Slow Motion Video", frame)

        if cv2.waitKey(100) & 0xFF == ord('q'):
            break

    video.release()
    cv2.destroyAllWindows()
```

Code (Fast Motion)

```
import cv2

video = cv2.VideoCapture("sample.mp4")

if not video.isOpened():
    print("Video not found!")
else:
    while True:
        ret, frame = video.read()

        if not ret:
            break

        # Resize frame
        frame = cv2.resize(frame, (640, 480))

        cv2.imshow("Fast Motion Video", frame)

        if cv2.waitKey(10) & 0xFF == ord('q'):
            break

    video.release()
    cv2.destroyAllWindows()
```

Input

Input Video: sample.mp4

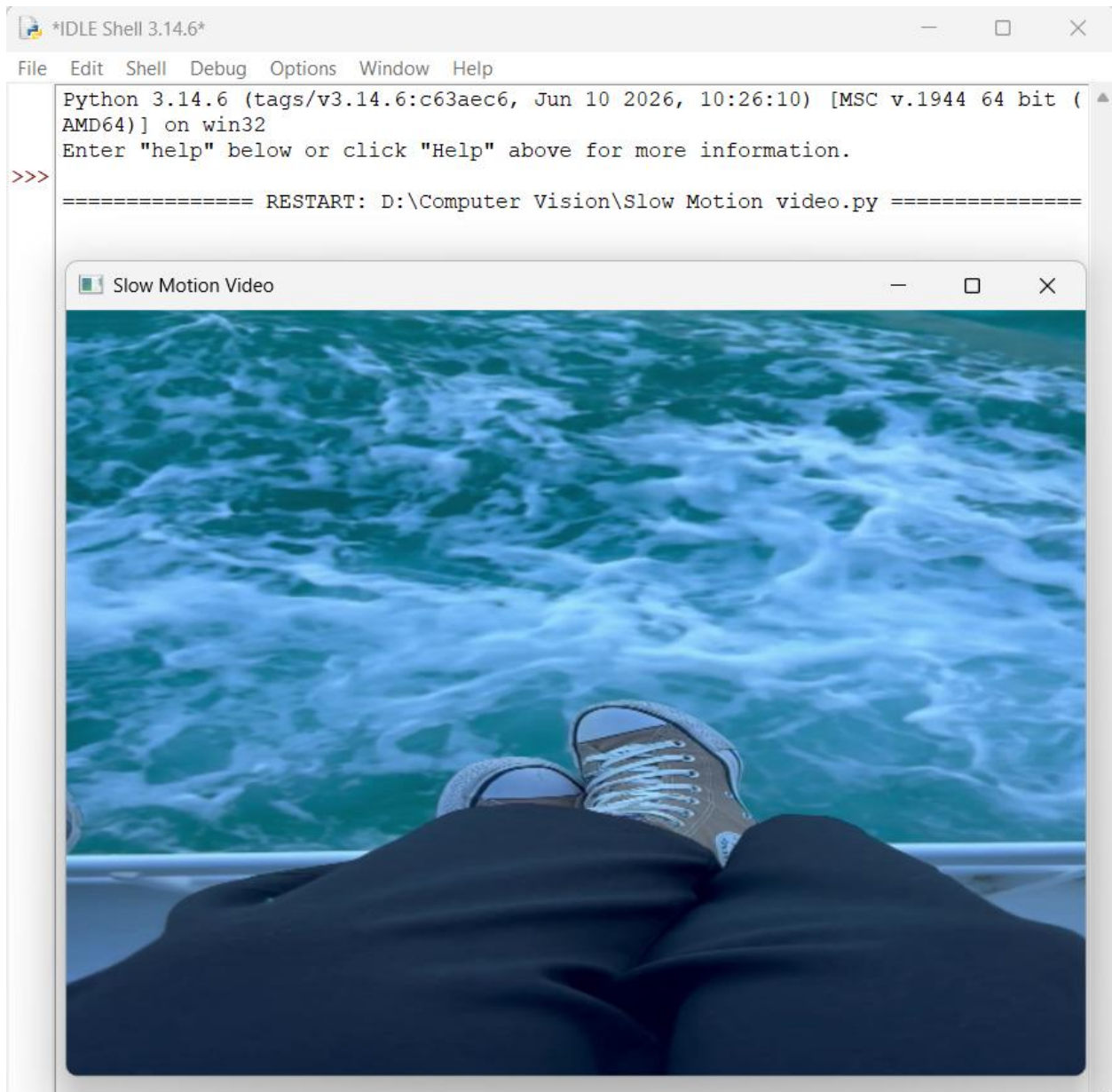
Output

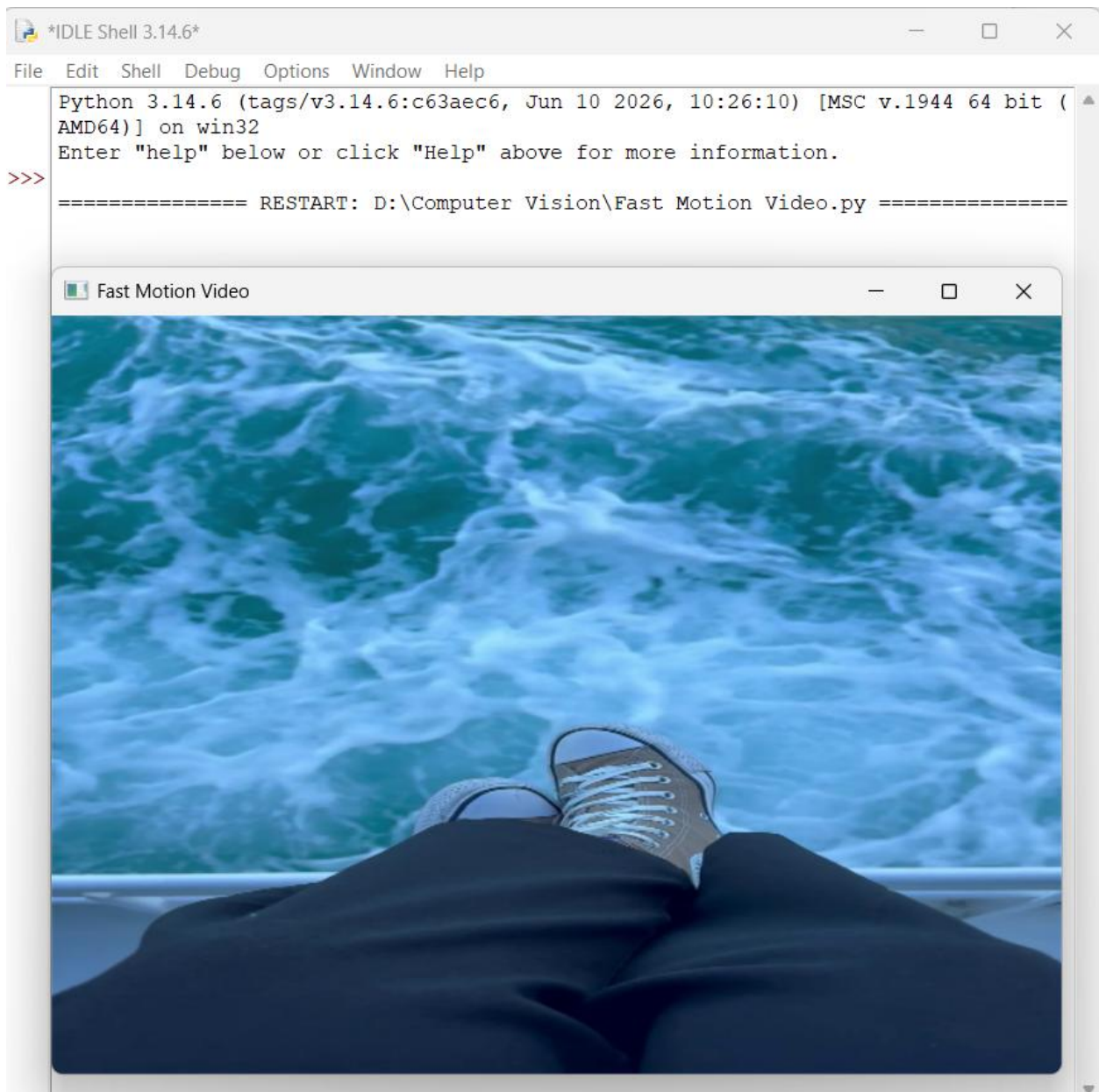
1. Slow Motion Video

- The video is displayed at a slower speed.

2. Fast Motion Video

- The video is displayed at a faster speed.





Result : The video was successfully read using OpenCV and displayed in both slow motion and fast motion.